

Software Dependability

Lab Session: JaCoCo, CodeCov, and PiTest

What are we going to do today?

Compute code coverage in Maven with JaCoCo

Compute code coverage in CI/CD

Apply mutation testing using PITest

Useful Links

JaCoCo

<https://www.jacoco.org/jacoco/>

<https://www.jacoco.org/jacoco/trunk/doc/maven.html>

Computing code coverage in IntelliJ IDEA

<https://www.jetbrains.com/help/idea/code-coverage.html>

Computing code coverage in CI/CD using CodeCov

<https://docs.codecov.com/docs/github-tutorial>

PiTest

<https://pitest.org/>

<https://github.com/pitest/pitest-junit5-plugin>

<https://www.baeldung.com/java-mutation-testing-with-pitest>

What's next?

You need to compute and analyze the code coverage of your project and conduct a mutation testing campaign for your project.